

# NEET Important Chapter – Electric Charges and Fields

## Electric Charges and Fields – An Important Concept of Physics

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The chapter Electric Charges and Fields notes down some of the most important concepts of electric charges and fields for NEET, JEE, and other competitive exams. In the Electric Charges and Fields chapter, students will study the basic properties of electric charge, conductors, and insulators, and some laws related to them.

The chapter on Electric Charges and Fields begins with the very basic definition of Coulomb's law, Electrostatic Forces, Dielectric, Electric Field, and what are different charge distributions helpful in studying the chapter ahead. These concepts will lead us to a very important part of the chapter, i.e. Vector notations of Electric Field, Electric Lines of Forces, Energy Density, and Electric Dipole.

The concepts mentioned above will help us in studying the chapter deeply along with the hands-on practice of many important numerical problems that will help us in acing the exams.

In this article, we will cover the important concepts and topics with the Electric Charges and Fields formulas as well as Electric Charges and Fields solutions for numericals that will help students to revise and boost their preparations for [NEET](#) and JEE.

## Important Topics of Thermal Physics

- Electric Charges and properties
- Coulomb's Law
- Electric Field
- Conductors and Insulators
- Dielectric
- Electric Lines of Forces
- Electric Flux
- Electric Dipole
- Gauss's Law

### Important Concepts of Thermal Physics

| Sl.No | Name of the Concept             | Key Points                                                                                                                                                                                                                                                 |
|-------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Electric Charges and properties | <ul style="list-style-type: none"> <li>• An invariant property which is related to matter producing and experiencing electric and magnetic effects.</li> <li>• Charges are of two types :</li> <li>• Positive charge</li> <li>• Negative charge</li> </ul> |

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| 2. | Properties of Electric Charges | <ul style="list-style-type: none"> <li>• Charge is a scalar quantity with value <math>e = 1.6 \times 10^{-19} \text{ C}</math> and SI Unit = Coulomb.</li> <li>• Opposite charges attract one other whereas similar charges repel against each other.</li> <li>• Charge is conserved i.e. the charge can neither be created nor be destroyed but it can be transferred from one body to the other.</li> <li>• <b>Quantisation of charge</b> - This property means that any body having charge existing on it will have an integral multiple of charge.<br/><math>q = ne</math> where <math>n</math> is an integer.</li> <li>• Charges are additive in nature which means charges follow algebraic law of addition.</li> </ul> |
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| 3. | Conductors and Insulators | <ul style="list-style-type: none"> <li>• Those materials which allow electricity to flow through them are known as conductors. Metals are very good conductors of electricity as they have free electrons associated with them which can move freely through the conductor.</li> <li>• <b>Insulators</b> :- The materials which do not allow electric charge to flow through them are called insulators.</li> <li>• For example Non metals, mica, wax, plastics, rubber all fall into the category of insulators.</li> </ul>                                    |
| 4. | Methods of Charging       | <ul style="list-style-type: none"> <li>• Mainly there are three methods of charging.</li> <li>• <b>By friction</b> : If we rub two suitable bodies such that one will be charged by +ve and another by -ve charge in equal amounts.</li> <li>• <b>By conduction</b> : When a neutral body is brought in contact with a charged body then that process is called charging by conduction.</li> <li>• <b>By induction</b> : If a neutral body without bringing it in contact with a charged body gets charged then that's called charging by induction.</li> </ul> |

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| 5. | Coulomb's Law  | <ul style="list-style-type: none"> <li>• Coulomb's law states :-</li> <li>• "The force of attraction or repulsion between any two point charges (<math>q_1</math> and <math>q_2</math>) separated at a finite distance "r" is directly proportional to the product of charges and inversely proportional to the square of distance between the charges. The direction of force is along the line joining the two charges.</li> <li>• <math>F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}</math></li> <li>• where <math>\epsilon_0</math> is the permittivity of free space.</li> </ul> |
| 6. | Dielectric     | <ul style="list-style-type: none"> <li>• A dielectric is an insulator which can be made a conductor when some kind of electric field is applied.</li> <li>• It is given by the formula <math>K = \frac{\epsilon}{\epsilon_0}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                            |
| 8. | Electric Field | <ul style="list-style-type: none"> <li>• Electric Field is considered as the space around an electric charge, where it exerts a force on another charge.</li> <li>• An electric charge produces an electric field around it so that it interacts with any other charges present there.</li> <li>• Unit of Electric Field is N/coulomb and is directed from positive charge to negative charge.</li> </ul>                                                                                                                                                                                 |

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| 9.  | Electric Lines Of Force               | <ul style="list-style-type: none"> <li>• These are imaginary lines of force around an electric region such that the tangent at any point on the lines of force gives the direction of the electric field at that point.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |
| 10. | Properties of Electric Lines of Force | <ul style="list-style-type: none"> <li>• The lines of force diverge from a positive charge and converge at a negative charge.</li> <li>• Two electric lines of force can never intersect each other as the intersection will give two different directions of force around a particular point which is not possible.</li> <li>• The Electric field lines don't make any loop like magnetic field lines.</li> <li>• The strength of the electric field can be determined from the electric lines of forces.</li> <li>• The close the field lines are to each other, the stronger the force is.</li> </ul> |



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| 11. | Electric Flux   | <ul style="list-style-type: none"> <li>• Electric flux is a measurement of the number of electric field lines passing through the closed surface and that closed surface is called Gaussian surface.</li> <li>• If the surface is a closed one enclosing some net charge, then the net number of lines going through the surface is proportional to net charge within the surface.</li> <li>• <math>\Phi = \frac{Q}{\epsilon_0}</math></li> <li>• The SI unit of Electric Flux is <math>\text{Nm}^2/\text{C}</math>.</li> </ul> |
| 12. | Gauss's Law     | <ul style="list-style-type: none"> <li>• Gauss's Law states the net electric flux through a closed surface in vacuum is equal to <math>1/\epsilon_0</math> times the net charge enclosed within the surface.</li> <li>• <math>\Phi = \frac{Q}{\epsilon_0}</math> or <math>\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}</math>.</li> </ul>                                                                                                                                                                                |
| 13. | Electric Dipole | <ul style="list-style-type: none"> <li>• An Electric Dipole is defined as a pair of equal and opposite charges separated by a distance <math>2a</math>.</li> <li>• The direction of the electric dipole is from negative potential to higher potential.</li> </ul>                                                                                                                                                                                                                                                              |

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| 14. | Physical Significance of Electric Dipole | <ul style="list-style-type: none"> <li>• Dipole moment helps to identify whether the given molecule is Polar or Nonpolar.</li> <li>• If the centres of positive and negative charges coincide or lie at the same place then they are said to be Nonpolar molecules with zero dipole moment.</li> <li>• If the dipole moment of some molecules is non zero even in the absence of electric field and centres of positive and negative charges don't coincide then they are called Polar Molecules.</li> </ul> |
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### List of Important Formulae of Electric Charges and Fields

| Sl. No | Name of the Concept                                                                       | Formulae                                                                                                                   |
|--------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1.     | Coulomb's Law                                                                             | $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$ <p>where <math>\epsilon_0</math> is the permittivity of free space.</p> |
| 2.     | Dielectric                                                                                | $K = \frac{\epsilon}{\epsilon_0}$                                                                                          |
| 3.     | Vector form of Coulomb's Law                                                              | $\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$                                                           |
| 4.     | Superposition Principle For Discrete Charge Distribution: Force Between Multiple Charges. | $\vec{F} = \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n$                                                                      |



|     |                                                                                                    |                                               |
|-----|----------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 5.  | Linear Charge Distribution                                                                         | —<br>Units of linear charge distribution = —  |
| 6.  | Surface Charge Distribution                                                                        | —<br>Units of surface charge distribution = — |
| 7.  | Volume Charge Distribution                                                                         | —<br>Units of volume charge distribution = —  |
| 8.  | Electric Field                                                                                     | —                                             |
| 9.  | Electric Field for Continuous Charge Distribution                                                  | — —                                           |
| 10. | A circular ring of radius R with uniformly distributed charge                                      | $E = \frac{\dots}{\dots}$                     |
| 11. | A circular disc of radius R with uniformly distributed charge with surface charge density $\sigma$ | $E = \frac{\dots}{\dots}$                     |
| 12. | An infinite sheet of uniformly distributed charges with surface charge density $\sigma$            | $E = \dots$                                   |

|     |                                                                                                  |                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 13. | Due to a spherical shell of uniformly distributed charges with surface charge density $\sigma$   | $= \frac{\sigma R^2}{2\epsilon_0} \text{ at } (x=R)$<br>$= \frac{\sigma R}{2\epsilon_0}$<br>$= 0 \text{ (} x < R \text{)}$ |
| 14. | Due to a solid non conducting sphere of uniformly distributed charges with charge density $\rho$ | $= 0$<br>$= \frac{\rho R^2}{2\epsilon_0}$<br>$= \frac{\rho R}{2\epsilon_0}$<br>$= \frac{\rho R}{2\epsilon_0}$              |
| 15. | Electric Flux                                                                                    | $=$                                                                                                                        |
| 16. | Gauss's Law                                                                                      | $=$ $=$ $=$                                                                                                                |
| 17. | Electric Dipole                                                                                  |                                                                                                                            |
| 18. | Electric Torque                                                                                  |                                                                                                                            |
| 19. | Work done in rotating the Electric Dipole                                                        |                                                                                                                            |

## Solved Examples of Electric Charges and Fields

1. An electric dipole with a dipole moment  $p$  C/m aligned at  $\theta$  with the direction of a uniform electric field of magnitude  $E$  N/C. Calculate the magnitude of torque acting on the dipole.

**Sol:**

**Given,**

Dipole moment  $p =$  C/m

$E =$  N/C and angle  $=$

Torque  $=$   $= pE \sin$

=

sin

=

Nm is the correct answer.

**Key point:** Here, the formula of torque due to an electric dipole can be used directly.

**2. Consider a uniform electric field  $E = \quad \text{N/C}$ .**

**(a) What is the flux of this field through a square of 10 cm on a side whose plane is parallel to the yz plane (b) What is the flux through the same square if the normal to its plane makes a  $60^\circ$  angle with the x-axis?**

**Sol:**

**Given,**

(a)  $E = \quad \text{N/C}$ , side of square = 10cm = 0.1m

Area of square =  $0.01\text{m}^2$ .

Since the plane of square is parallel to the yz plane and the area vector is also in this plane so the angle between  $E$  and area vector is zero.

=

= = =

=

**=  $30 \text{ Nm}^2/\text{C}$ .**

(b) =

= = =

=

**=  $15 \text{ Nm}^2/\text{C}$**

**Key point:** Here, the formula of electric flux can be used directly.

### Previous Year Questions of Electric Charges and Fields

1. Two parallel infinite line charges with linear charge densities  $+ \lambda \text{ C/m}$  and  $- \lambda \text{ C/m}$  are placed at a distance of  $2R$  in free space. What is an electric charge field midway between the two line charges? (NEET 2019)

- a. Zero
- b.  $\frac{\lambda}{2R} \text{ N/C}$
- c.  $\frac{\lambda}{R} \text{ N/C}$
- d.  $\frac{\lambda}{2R^2} \text{ N/C}$

**Sol:**

Given,

As one can clearly see the linear charge distribution of parallel plates and the direction of  $E$  by both the plates on each other so the electric field due to plate 1 having linear charge density  $-\sigma$  is

$$E_1 = \frac{\sigma}{2\epsilon_0} \text{ N/C}$$

Similarly the electric field due to plate 2 having linear charge density  $+\sigma$  is

$$E_2 = \frac{\sigma}{2\epsilon_0} \text{ N/C}$$

As the Electric Field follows the vector superposition principle so the net  $E = E_1 + E_2$

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

**Which is option (c)**

**Trick:** The relationship between the Electric Field having \_\_\_\_\_ as linear charge density is  
= \_\_\_\_\_.

**2. Two point charges A and B, having charges Q and -Q respectively, are placed at a certain distance apart and the force acting between them is F. If 25% charge of A is transferred to B, then the force between charges becomes: (NEET 2019)**

**01. F**

**02.  $9F/16$**

**03.  $16F/9$**

**04.  $4F/3$**

**Sol:**

Given,

We have been provided two point charges A and B having charges +Q and -Q separated by distance r having force F between them.

Hence  $F = \text{_____}$



Now 25% of the charge is transferred from A to B then the charge left on A

$$Q_A = Q - \frac{Q}{4} = \frac{3Q}{4}$$

Similarly the charge on B now is

$$Q_B = \frac{Q}{4}$$

So the new forces due to  $Q_A$  and  $Q_B$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_A Q_B}{r^2}$$

$$F = \frac{9}{16} F$$

**Which is option (b)**

**Trick:** Here, the formula of electrostatic force between two charge particles can be used directly.

## Practice Questions

1. A Conducting sphere of radius 10cm has an unknown Charge. If the Electric Field at a Point 20cm from the center of the sphere of magnitude  $1.5 \times 10^3 \text{ N/C}$  is directed radially inwards, what is the net charge on the sphere?

(Ans: 6.67nC)

2. Four charges  $q_A = 2 \text{ microcoulomb}$ ,  $q_B = -5 \text{ microcoulomb}$ ,  $q_C = 2 \text{ microcoulomb}$ ,  $q_D = -5 \text{ microcoulomb}$  are located at the corners of a square ABCD of side 10cm. What is the force on a charge of 1micro microcoulomb placed at the centre of a square?

(Ans: zero)

## Conclusion

We conclude that the chapter electric charges and fields has many important concepts. In this article, we covered Electric Flux, Gauss's Law and all other important concepts from the chapter along with solved and previous year questions. Students can test their knowledge with the help of practice questions.

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